INCH-POUND

MIL-I-24768/4
25 February 1992

SUPERSEDING
MIL-P-24364/1
11 July 1969

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, GLASS-MAT, POLYESTER-RESIN (GPO N-1)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a mat of random-laid glass fibers which is saturated with a polyester-resin combined with suitable fillers, cured by heat and pressure and processed to conform to this specification.

Intended use: The laminated material covered by this specification is a rigid plastic material intended for general-purpose electrical equipment applications requiring good dielectric properties.

Forms: Sheets, are the only recommended form of laminated material covered by this specification.

Temperature classification: The laminated material covered by this specification has a temperature classification of class 130, class 155, or class 180.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 105, and an electrical strength test index of 105 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

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Performance requirements: Performance requirements for laminated sheets shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Volume resistivity
b. Surface resistance
c. Dielectric strength
d. Specific gravity
e. Compressive strength
f. Flammability.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/4-S-1	0.031	0.008
M24768/4-S-2	0.062	0.008
M24768/4-S-3	0.094	0.009
M24768/4-S-4	0.125	0.010
M24768/4-S-5	0.156	0.011
M24768/4-S-6	0.188	0.012
M24768/4-S-7	0.219	0.014
M24768/4-S-8	0.250	0.015
M24768/4-S-9	0.312	0.018
M24768/4-S-10	0.375	0.020
M24768/4-S-11	0.438	0.022
M24768/4-S-12	0.500	0.024
M24768/4-S-13	0.625	0.027
M24768/4-S-14	0.750	0.029
M24768/4-S-15	0.875	0.031
M24768/4-S-16	1.000	0.033
M24768/4-S-17	1.125	0.035
M24768/4-S-18	1.250	0.037
M24768/4-S-19	1.375	0.039
M24768/4-S-20	1.500	0.041
M24768/4-S-21	1.625	0.043
M24768/4-S-22	1.750	0.045
M24768/4-S-23	1.875	0.047
M24768/4-S-24	2.000	0.049
M24768/4-S-25	–[1]	–[2]

[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/4-S-25-0.8125.

[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

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TABLE V. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.062	0.125	0.250	0.500	1.000	1.001 – 2.000
Dielectric breakdown voltage	A D-48/50 D-336/50	4.6.4	Min. kV	40.0 15.0 15.0	40.0 15.0 15.0	40.0 15.0 15.0	40.0 15.0 15.0	15.0	15.0
Dielectric constant									
60 Hz:	A D-24/23	4.6.6	Max.		6.0 6.5				
1 MHz:	A D-24/23				5.0 6.0				
Dissipation factor									
60 Hz:	A D-24/23	4.6.6	Max.		0.10 0.15				
1 MHz:	A D-24/23				0.03 0.06				
Volume resistivity	C-96/35/90	4.6.7	Min. meghom cm						
Surface resistance	C-96/35/90	4.6.8	Min. megohm						
Arc resistance	A D-48/50	4.6.9	Min. seconds	100 100	100 100	100 100	100 100		
Tracking resistance	A	4.6.10	Min. minutes						

See footnotes at the end of table.

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TABLE IV. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]					
				0.062	0.125	0.250	0.500	1.000	1.001–2.000
Impact strength	E-48/50	4.6.11	Min. ft–lbs. per inch	8.0	8.0	8.0	8.0		
Flexural strength	A	4.6.12	Min. psi	18,000	18,000	18,000	18,000	18,000	18,000
Thermal endurance[2]									
Flexural strength:	E-1/130:T-130	4.6.13	Min. psi	6,000	6,000	6,000	6,000	6,000	6,000
	E-1/155:T-155			9,000	9,000	9,000	9,000	9,000	9,000
	E-1/180:T-180			9,000	9,000	9,000	9,000	9,000	9,000
Bonding strength	A	4.6.15	Min. pounds				850	850	850
	D-48/50						800	800	800
Water absorption	D_1-24/23	4.6.17	Max. percent	0.9	0.6		0.35		
Silicone content[3]	E-168/185	4.6.18	Max. ppm	25.0	25.0	25.0	25.0	25.0	25.0
Flame resistance									
Ignition time:	A	4.6.19	Min. sec.						
Burning time:			Max. sec.						

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Select the elevated temperature flexural test which corresponds to the desire temperature classification of laminated material.
[3]Specimens of the same material shall be built up to 0.250 inch.

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User activities:
Army – ME
Navy – MC, AS

Preparing activity:
Navy – SH
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